Constrained Optimization
Optimal Drug Injection for Cancer Treatment (4h)

Objective

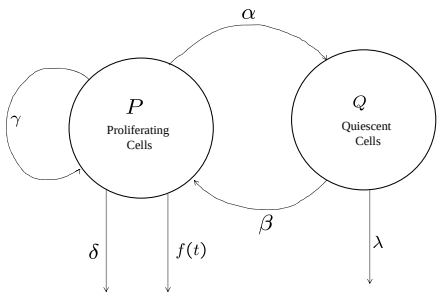
Solve an example of constrained optimal control problem. Learn how to define cost functions that are related to dynamic systems modelled by Ordinary Differential Equations. Use a non-linear constrained optimizer (FMINCON).
A two-pages report must be posted in a pdf format to the dedicated Chamilo folder at the end of the computer session.

Problem Statement

The basic dilemma in cancer treatment by chemotherapy is to inject enough drug to reduce the population of proliferating cells while preserving the health of the patient. This is because drug also affects the healthy cells performance.

There are many models that intend to describe these effects. The following is one of the simplest one (see also the right Figure for a schematic view of the population model) :

$$\dot{P}(t) = [\gamma - \delta - \alpha - u(t)]P(t) + \beta Q(t) \quad (1)$$
$$\dot{Q}(t) = \alpha P(t) - (\lambda + \beta)Q(t) \quad (2)$$



Schematic view of the population model for cell-cycle specific chemotherapy

where :
- P : the number of proliferating cells
- Q : the of Quiescent (healthy) cells
- γ : the proliferating cells growth rate
- δ : the proliferating cells natural death rate
- α : the transition rate from proliferating state to the quiescent state
- β : the transition rate from quiescent to proliferating state
- λ : the cells differentiation rate (mature bone-marrow cells leaving the bone marrow to enter blood stream)
- $f(t)$ is the drug control variable that acts on the proliferating cells.

Note that if too much drug is injected P decrease too fast and leave Q without any positive term. This mechanism lead to the negative side effect described above. To avoid that, the following optimization problem is defined for the treatment of duration T :

$$\min_{u(\cdot)} \Big[P(T) + \mu \cdot \int_0^T u(\tau) d\tau \Big]$$
$$\text{under the constraints } u(t) \in [0,1] \text{ and } P(t) + Q(t) \geq \rho \text{ for all } t \in [0,T] \quad (3)$$

where $P(T)$ is the number of proliferating cells at the end of the treatment and ρ is a lower bound on $P + Q$ as the condition $(\forall t) P(t) + Q(t) \geq \rho$ guarantees that the patient is not too diminished by the treatment.

The aim of this project is to provide a solution to this constrained optimization problem.

In this project, the numerical values proposed by [1] are used :

$$\gamma = 1.47 \ ; \ \alpha = 5.64 \ ; \ \lambda = 0.16 \ ; \ \delta = 0 \ ; \ \beta = 0.48 \ ; \ T = 30 \ ; \ \rho = 0.35 \ ; \ \mu = 10^{-3}$$

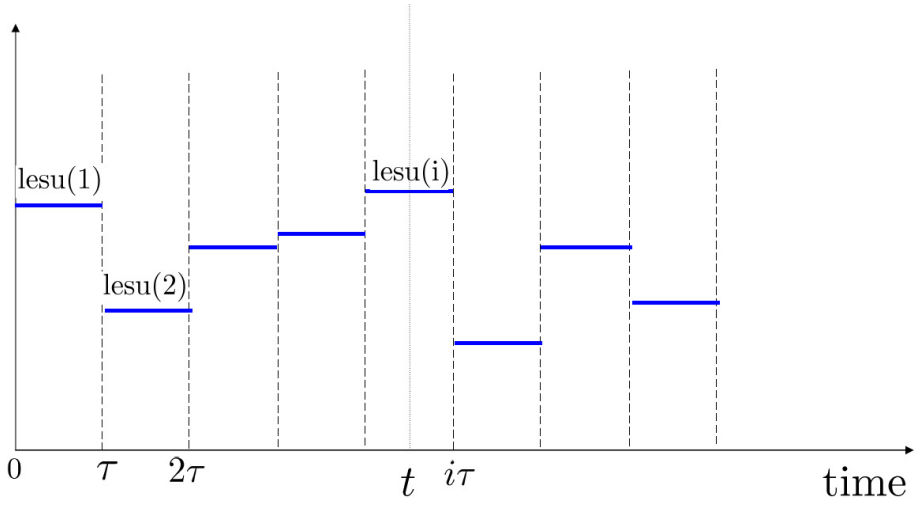
Questions

1. Write a function that implements the ordinary differential equation (1)-(2) for use with the MATLAB ODE solvers according to the following syntaxe :

   xdot=my_ode(t,x,lesu)

   where *lesu* is a vector of N successive control actions defining a piecewise constant control by :

   $$u(t) = lesu(i) \qquad \text{where } i \text{ is such that } t \in [(i-1)\tfrac{T}{N}, i\tfrac{T}{N}[$$

   [Hint : Use global variable status for the model coefficients α, β, etc].

2. Using above function, write a function that implement the cost function in (3) for use with the MATLAB optimizer FMINCON, namely

   J=cost(lesu)

   [Hint : here again, the initial state x_0 must be fed as a global variable as well as the treatment duration T, etc.]

3. write a function that compute the constraints defined in the optimization problem (3) according to the syntaxe required by FMINCON

4. Compute the optimal Drug injection strategies for $P(0) = Q(0) = 0.5$ and the following different values of $N \in \{2, 5, 10\}$ and starting from the *initial guess lesu=(0,0,...,0)*. For each case, plot the resulting optimal solution (evolution of P, Q, $P + Q$ and u).

5. Repeat the preceding question using randomly generated initial guesses for *lesu*.

6. Repeat the preceding question using successively $\mu = 0.01$ and $\mu = 0.1$. Explain the results.

7. Give all the observations you find relevant concerning your experimentation.

references

[1] K. R. Fister and J. C. Panetta. *Optimal control applied to cell-cycle specific cancer chemotherapy. SIAM Journal on Applied Mathematics, 60 (3) :1059-1072, 2002.*